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Predict the Medal Data of Games of the XXXIV Olympiad

Summary

The first part of the paper focuses on medal count predictions. The model leverages historical Olympic data from 1900 to 2024 to predict medal counts for future Games. By using machine learning techniques such as **linear regression** and **neural networks**, it offers reliable predictions for major Olympic nations and provides **confidence intervals** to account for prediction uncertainty. The accuracy of predictions is particularly strong for countries with consistent participation and medal histories, while countries with limited data may show more variation.

The second section, **first-time medal predictions**, examines countries that have not won medals by 2024 but are likely to earn their first medals in 2028. The **random forest model** identifies 18 countries as potential first-time medalists based on various factors, including changes in national participation, sport preferences, and emerging athlete talent.

The third part of the paper explores the relationship between Olympic events and medal outcomes. The model uses event characteristics—such as the number of events, sport type, and country participation—alongside a fully connected **neural network (FCN)** to predict medal counts. Starting with a **linear regression** model to establish a baseline, it links event features with medal outcomes. The model is then refined with a **neural network** to capture more complex patterns. This approach helps predict how specific events and their characteristics affect a country's overall medal performance.

The fourth part of the paper introduces the "**great coach**" effect, examining how coaching changes impact athletic performance. The model quantifies the relationship between coaching transitions and medal outcomes, showing that leadership can significantly influence a country's success at the Olympics.

In conclusion, the paper demonstrates that combining historical data, event features, coaching effects, and other variables creates a robust and flexible tool for predicting Olympic outcomes. While the model's predictions are influenced by data quality and external factors, its ability to forecast medal counts and identify future medalists provides valuable insights for Olympic committees and sports analysts.

Keywords: Oylimpic Medal; Linear Regression; Neural Network; Random Forest

Contents

1 Introduction

1.1 Problem Background

The Olympic Games are a highly anticipated global event where nations compete for gold, silver, and bronze medals. The final medal table, showing the total number of medals won by each country, is closely watched, with countries striving for top rankings. For example, in the 2024 Paris Olympics, the United States led with 126 total medals and tied with China for golds (40). Countries like France and Great Britain ranked high in total medals, despite fewer golds.

Paris Olympics 2024 medal table

	Country	Gold	Silver	Bronze	Total
1	United States	40	44	42	126
2	China	40	27	24	91
3	Japan	20	12	13	45
4	Australia	18	19	16	53
5	France	16	26	22	64
6	Netherlands	15	7	12	34
7	Great Britain	14	22	29	65
8	South Korea	13	9	10	32
9	Italy	12	13	15	40
10	Germany	12	13	8	33

Figure 1: The Rank of Paris Olympics 2024

pedicting medal counts for future Olympics, such as the 2028 Los Angeles Games, is a complex challenge. Medal projections are often based on recent athlete performances, but several factors influence a country's success, including the number and types of events, athlete skill, and external factors like coaching.

The goal is to develop a model to predict medal counts for the 2028 Olympics using historical data. Key objectives include:

- Projections for 2028: Forecasting medal counts for each country, with prediction intervals to account for uncertainty.
- First-Time Medalists: Estimating the likelihood of countries earning their first medal.
- Sport-Specific Trends: Analyzing how event changes impact national success in specific sports.
- Coaching Impact: Examining how top coaches influence medal counts and identifying sports where investment in coaching could yield significant results.

1.2 Restatement of the Problem

The goal of this problem is to develop a mathematical model that can predict Olympic medal counts for countries based on historical data from past summer Olympic Games. The model should specifically address the following key objectives:

- **Model Development:**

✓Create a model to predict the number of gold and total medals each country will earn, using historical data of previous Olympic Games.
✓Estimate the uncertainty and precision of the predictions, and evaluate the model's performance.

- **Projections for the 2028 Los Angeles Olympics:**

✓Using the developed model, project the medal table for the 2028 Los Angeles Olympics, providing prediction intervals for each country's medal count.
✓Identify which countries are likely to improve or perform worse compared to their 2024 results.

- **First-Time Medalists:**

✓Include countries that have not yet won Olympic medals, and predict how many will earn their first medal in the next Olympics.
✓Provide estimates of the odds for these countries earning their first medal.

- **Impact of Events on Medal Counts:**

✓Analyze how the number and types of events at each Olympics influence the medal counts for countries.
✓Explore the relationship between different sports and the performance of various countries, identifying the key sports that contribute most to a country's medal count.

- **Great Coach Effect:**

✓Investigate whether the performance of countries can be influenced by the presence of a "great coach" effect, where a coach's expertise helps multiple countries (e.g., Lang Ping with volleyball and Béla Károlyi with gymnastics).
✓Estimate the impact of the "great coach" effect on medal counts for specific countries and sports.

- **Insights for Olympic Committees:**

✓Provide any original insights revealed by the model regarding medal distributions and trends.
✓Suggest how these insights can help national Olympic committees plan their strategies for the upcoming Olympics, including investment in coaching and athlete preparation.

This approach aims to build a predictive model of Olympic medal outcomes, evaluate the role of event types and coaching strategies, and offer actionable insights to improve future Olympic performances for countries around the world.

1.3 Our Work

We conducted extensive data cleaning and exploration to address issues like missing data and normalization across different Olympic editions. Building on this, we developed two key models: the Medal Rank Prediction model, which predicts country rankings in total medals and Gold medals by factoring in variables like events, population, and sport specialization using machine learning techniques, and the First Medal Prediction model, which estimates the likelihood of countries winning their first medals, taking into account athlete preparation, infrastructure, and regional trends. Additionally, we analyzed the impact of event types and coaching changes on performance, such as examining how coaching transitions impact can influence medal outcomes. Our model incorporates uncertainty quantification and validation using historical data to ensure its reliability. Overall, our work provides a comprehensive framework for predicting the 2028 Olympic medal table and offers valuable insights into strategies that may shape future Games.

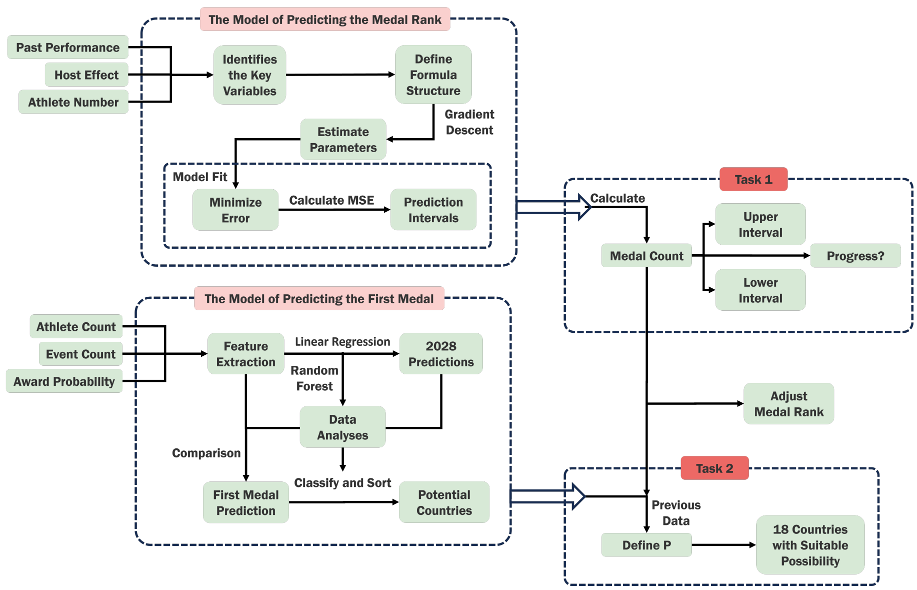


Figure 2: The Mind Map of Our Work

2 Assumptions And Notations

2.1 Assumptions

- Assumption 1: A country's performance is correlated with its performance in previous years

This assumption is based on the idea that a country's Olympic performance tends to follow a certain trend over time. Historical data often shows that countries with a strong track record in the Olympics are likely to continue performing well due to factors like sustained investment in sports development, training infrastructure, and experience gained from previous competitions. Past performances can also reflect the effectiveness of a country's sporting policies and the depth of its athletic talent pool, which often leads to consistent results in future Olympics.

- Assumption 2: The country's performance is positively correlated with whether it is the host nation

This assumption suggests that being the host nation can provide a significant advantage in terms of both psychological and logistical factors. Hosting the Olympics often boosts national pride and provides athletes with a sense of support from their home country. Additionally, home teams may benefit from reduced travel fatigue, better acclimatization to the local environment, and a higher level of crowd support, all of which can enhance performance. This phenomenon is often referred to as the "home advantage," where host nations tend to perform better than usual.

- Assumption 3: The country's performance is positively correlated with the number of participants it sends to the games

The assumption here is that countries with larger delegations are likely to perform better overall in terms of medal counts. A larger team increases the chances of success simply because there are more athletes competing across various sports and events. Additionally, a larger number of participants means more specialized athletes who can excel in different disciplines, contributing to an overall higher medal tally. Countries that send a broader representation of athletes have a higher likelihood of securing medals across a diverse range of events.

2.2 Notations

In our paper, we have adopted the nomenclature presented in Table 1 for model construction. Any additional symbol which is not frequently used will be introduced once they are utilized in the following text. All variables are targeted at a certain set, and the training data for the model also includes data of an entire set. Definitions without these words means value at current moment.

Symbol	Definition
s	Number of the Athletes from A Country
p	The Predicted Medal Count for The Given Year
i	The Name of A Team
P_i	The Possibility of A Country to Win The First Medal
Pe_i	The Athlete's Current Performanc

Table 1: Notations

2.3 Data Cleaning

2.3.1 Missing and Abnormal Data

We addressed several issues in the dataset:

- **Missing Medal Counts**: Replaced "NA" with 0 for countries without medals in a specific year to maintain consistency.

- **Inconsistent Country Names**: Standardized country names according to IOC guidelines and merged entries like "Germany-1" to "Germany".

- **Incorrect Total Medal Counts**: Recalculated total medals by summing Gold, Silver, and Bronze, correcting discrepancies where needed.

2.3.2 Different Types of Data

- **Numerical Data**: Ensured consistency by recalculating total medals and correcting any inconsistencies in the data.

- **Categorical Data**: Standardized country and event names to match IOC classifications.

2.3.3 Data Normalization and Transformation

- **Event Count Adjustment**: Normalized medal counts by adjusting for the number of events in each Olympic year.

- **Medal Performance Normalization**: Adjusted medal counts by country size and historical performance for fairer comparisons.

- **Coaching Effects**: Incorporated assumptions on the influence of top coaches, such as in volleyball or gymnastics, to inform predictions.

3 Calculating and Optimizing the Model

3.1 The Model of Predicting the Medal Rank

To derive the formula for predicting a country's medal count at the Olympics, we start by recognizing that several factors likely influence a country's performance. The core idea is to look at historical data and identify key variables that might correlate with the number of medals won. We also acknowledge the inherent randomness in the data, which means our model may not provide perfect predictions but can offer reasonable estimates in specific cases.

Step 1: Identify Variables

- **Past Performance:** We hypothesize that a country's medal performance in past Olympics (specifically the last three Olympics) may influence its performance in the current one. Hence, the number of medals won in the previous three Olympics is included as a key factor.

- **Host Country:** Whether a country is the host of the Olympics is another important factor. Host countries often perform better due to the home advantage, so this is included as a binary variable (1 for host, 0 for non-host).

- **Athletes Sent:** The number of athletes a country sends to the Olympics could also be a significant predictor. Generally, more athletes equate to more opportunities for medals, so this number will be factored in as well.

Step 2: Structuring the Formula

We aim to construct a formula that integrates these variables:

- Let p_n represent the medal count prediction for a given year.
- We know from the assumptions above that a country's past performance plays a role, so we incorporate the medal counts from the last three Olympics, denoted as p_{n-1}, p_{n-2}, and p_{n-3}. These will be weighted by parameters a, b, and c, respectively, to capture their relative influence.
- The γ^i term adjusts for the effect of being the host country. γ is a parameter that will scale the influence of this variable, and i is 1 if the country is hosting the Olympics and 0 if not.
- n^d represents the influence of the number of athletes sent to the Olympics. We hypothesize that sending more athletes should increase the number of medals, so this term will scale according to n, the number of athletes, raised to the power of d, another parameter to tune.

Thus, the formula becomes:

$$p_n = (a \cdot p_{n-1} + b \cdot p_{n-2} + c \cdot p_{n-3}) \cdot \gamma^i \cdot s^d$$

Where:

- a, b, c, γ, and d are the parameters to be determined using historical data.
- p_n is the predicted medal count for the given year.
- i is 1 if the country is hosting the Olympics, 0 otherwise.
- s is the number of athletes sent to the Olympics.

Step 3: Estimating Parameters and Model Fit

To estimate the parameters a, b, c, γ, and d, we use a regression-based method, such as gradient descent. This approach allows us to fit the model to historical data and find the best values for the parameters that minimize the error between predicted and actual medal counts. Due to the inherent randomness in the Olympic data, the model's fit is not perfect. However, the results for certain countries with more complete and reliable data are reasonably accurate, demonstrating that the model can capture the general trends in medal counts. The model has four degrees of freedom, which restricts the complexity of the fitting process but still allows for meaningful predictions. In cases where the data is more robust and consistent, the model can closely mimic historical medal trends, even if the fit is not exact for every country.

Step 4: Confidence Intervals

Given the variability in the data, we calculate prediction intervals to express the uncertainty in the model's estimates. To do this, we first compute the mean squared error (MSE) from the model's predictions. From this, we derive the standard deviation of the residuals and apply a 95% confidence interval, using 1.96 times the standard deviation to calculate the range within which the true medal count is likely to fall. This provides a measure of uncertainty and allows us to better understand the reliability of our predictions.

Conclusion

By structuring the problem in this way, we create a model that takes into account historical performance, the home advantage of being the host, and the number of athletes sent to the Olympics. These factors combined give us a prediction of the medal count for a given country in a future Olympic Games. While the model's fit may vary depending on data quality and the randomness in Olympic performance, it can provide reasonable estimates and useful insights, especially for countries with more complete and reliable data. The use of confidence intervals further enhances the robustness of the predictions, making this model a valuable tool for Olympic medal count forecasting.

3.2 The Model of Predicting the First Medal

3.2.1 Thought Process

1. **Initial Data Filtering:** The first step is to identify countries that had no medals by 2020, but won at least one medal by 2024. This is important because it highlights nations that have undergone significant improvement or breakthrough performances.

2. **Feature Extraction:** For the selected countries, several key features are extracted:

   - **Athlete count:** The number of athletes each country has sent to the Olympics.
   - **Event count:** The number of events in which each country participated.
   - **Award probability:** This could be calculated based on past performances or other relevant metrics such as growth in athlete count and event participation.

3. **Linear Regression for Predicting 2028 Data:** To forecast the number of athletes and events for each country in 2028, we first apply a linear regression model based on historical data. Specifically, we use the data from previous Olympics to predict how the number of athletes and events will change by 2028. This provides us with the necessary predicted data points for the countries that did not win medals by 2024. The linear regression model helps us capture the general trends in participation growth over time.

4. **Machine Learning Model:** A machine learning model is applied to predict which countries that have not won medals by 2024 might win their first medal by 2028. Historical data and trends (from 2020 to 2024) are used as inputs, with a particular focus on the countries' participation growth in terms of athletes and events.

   The choice of **Random Forest** for this prediction is motivated by its ability to handle non-linear relationships and complex feature interactions, which are inherent in the Olympic data. For instance, the relationship between the number of athletes sent, the number of events participated in, and the likelihood of winning a medal is not straightforward. Random Forest can capture these interactions effectively without making assumptions about the data distribution. Moreover, it is robust against overfitting, which is crucial given that some countries may have limited historical data or noisy performance patterns. The model can also handle high-dimensional data, selecting the most relevant features, and offers interpretability by identifying the importance of each feature in making predictions.

5. **Data Comparison and Prediction:** After predicting the 2028 data for each country (which has been done previously with linear regression), the predicted data for countries without medals by 2024 is compared. The idea is to identify countries that show similar trends to those that won their first medal by 2024.

6. **Goal:** The goal is to predict, based on historical growth and participation, which countries will potentially win their first medal in 2028. By comparing the data of countries that have recently shown growth in their Olympic participation and performance, patterns can be identified to help predict future success.

3.2.2 Formula Structure

Based on the steps outlined, the mathematical model can be formulated as follows:

1. Growth Rate Calculation

For each country i, the growth rates between 2020 and 2024 are computed for key metrics:

- **Athlete growth rate:**

$$\text{growth\_rate\_athlete}_i = \frac{\text{Athlete\_Count\_2024}_i - \text{Athlete\_Count\_2020}_i}{\text{Athlete\_Count\_2020}_i}$$

- **Event growth rate:**

$$\text{growth\_rate\_event}_i = \frac{\text{Event\_Count\_2024}_i - \text{Event\_Count\_2020}_i}{\text{Event\_Count\_2020}_i}$$

- **Medal growth rate:** (if applicable)

$$\text{growth\_rate\_medal}_i = \frac{\text{Total\_2024}_i - \text{Total\_2020}_i}{\text{Total\_2020}_i}$$

2. Linear Regression for Predicting 2028 Data

We use linear regression to predict the number of athletes and events for each country in 2028:

- **Features:** For each country i, the features used to predict 2028 data are based on historical participation trends:

$$X_i = [\text{Athlete\_Count\_2020}_i, \text{Event\_Count\_2020}_i]$$

- **Target:** The target is the predicted values for 2028:

$$\text{Predicted\_Athlete\_Count\_2028}_i, \text{Predicted\_Event\_Count\_2028}_i$$

The linear regression model is trained on historical data and used to predict the future values for each country.

3. Machine Learning Model

Using the growth rates from the linear regression model, we train a machine learning model (**Random Forest Classifier**) to predict whether a country will win its first medal in 2028. The features used for prediction are the growth rates for athletes and events:

- **Features:**
$$X_i = [\text{growth\_rate\_athlete}_i, \text{growth\_rate\_event}_i]$$
- **Target:**
$$y_i = \begin{cases} 1 & \text{if the country wins a medal in 2028} \\ 0 & \text{if the country does not win a medal in 2028} \end{cases}$$

The classifier learns from this data and outputs a predicted probability P_i for each country:

$$P(\text{Win Medal in 2028})_i = \text{RandomForestClassifier}(X_i)$$

The classifier predicts whether the growth rates are sufficient for a country to break through and win its first medal.

4. Final Prediction

After training the model, we apply it to countries that have not won medals by 2024. The output is a **probability** P_i that each country will win a medal in 2028.

5. Threshold for Prediction

If the probability P_i exceeds a threshold (e.g., $P_i > 0.5$), the country is classified as likely to win a medal in 2028.

3.3 The Model Between Events And Medals

3.3.1 Model Overview and Structure

The model utilizes a fully connected neural network (FCN) to model the relationship between input event characteristics and the medal outcomes (gold, silver, bronze, total). The input consists of various features of Olympic events (such as type, number of events, and countries participating). The output will be the predicted medal counts for each country based on these events.

1.Data Selection for Training

The model will be trained using data from 1980 and onwards, focusing on 7 representative countries. This data captures a broad range of Olympic event performances, providing a diverse representation of how countries perform across various Olympic events.

2.Linear Regression Fitting

We start with linear regression to model the relationship between event characteristics and medal outcomes. In this context, the dependent variable y represents the medal counts (e.g., total medals won by a country in a given Olympics), and the independent variables x represent event features (such as the number of events, event types, and other relevant factors). The linear regression equation is given by:

$$y = wx + b$$

where:

- y is the predicted medal count,
- x represents the input features (event-related variables),
- w is the weight (coefficient) that reflects the influence of the event type on the medal count,
- b is the bias term.

3.Weights and Biases

During training, the model learns the values of the weights (w) and the bias (b) that minimize the difference between the predicted and actual medal counts. The magnitude of each weight (w) indicates how strongly each event characteristic (e.g., event type) influences the medal outcomes. The bias (b) accounts for other factors that might affect performance, such as a baseline medal count.

4.Training Process

The model is trained using stochastic gradient descent (SGD), where the weights and biases are iteratively adjusted to minimize the error between predicted and actual values. A small learning rate is chosen to prevent gradient explosion during the optimization process, ensuring that the training remains stable and converges to a minimum error. Linear regression provides a baseline model for predicting medal counts, which is further refined by the neural network to capture more complex relationships.

3.3.2 Evaluation and Prediction

1.Model Evaluation

After training, the model's performance is evaluated using the Mean Squared Error (MSE), which should be small if the model is accurately predicting the medal counts. The linear regression model offers a simple yet interpretable baseline, while the neural network can refine this prediction by capturing more complex relationships.

2.Prediction and Generalization

Once trained, the model can predict medal outcomes for any given Olympic event setup. The input and output structures during prediction will be consistent with the training data, allowing the model to predict the number of medals each country might win based on event characteristics.

3.Final Model

The final model will be capable of predicting future Olympic Games medal outcomes, considering the number and types of events as well as country-specific performance trends. The trained weights and biases will be used to generalize the results for different event scenarios.

3.3.3 Conclusion

By using linear regression as a foundational model and applying stochastic gradient descent for optimization, the model learns the relationships between event characteristics and medal outcomes. The linear equation $y = wx + b$ provides an interpretable fit for understanding how event characteristics affect medal counts, while the neural network refines this fit to improve accuracy and prediction.

3.4 The Model of Great Coach Effect

3.4.1 Introduction and Proposed Model

The existence and influence of the "great coach" effect on athlete performance is a topic of considerable interest in sports analysis. Traditional methods often rely on changes in athlete performance to determine the presence of this effect. However, due to the complexity of this phenomenon and the multiple factors at play, directly linking performance changes to the "great coach" effect can be challenging. To address this, we propose a new model that quantifies the impact of this effect by simultaneously considering changes in coaching and athlete performance.

1.Regression Model for Great Coach Effect

To examine the relationship between coaching changes and athlete performance, we propose the following regression model:

$$Pe_i = \beta_0 + \beta_1 X_i + \beta_2 Y_i + \beta_3 X_i Y_i$$

Where:

- Pe_i represents the athlete's current performance (which can be measured by their ranking).
- X_i is a binary indicator variable for coaching change, where $X_i = 1$ if there was a coaching change and $X_i = 0$ if there was no change.
- Y_i is the change in performance during the period after the new coach took over, reflecting improvements or declines in athlete performance during this time.

The coefficient β_3 represents the interaction effect between coaching changes and performance changes, and this is the key parameter for estimating the magnitude of the "great coach" effect.

2.Categories of Performance Change

The value of Y_i can have three possible outcomes:

- $Y_i = 0$: No significant change in performance, indicating no change in ranking or medals.

- $Y_i = 1$: Performance improvement, indicating an increase in rankings or medals.
- $Y_i = -1$: Performance decline, indicating a decrease in rankings or medals.

By incorporating these outcomes, we can capture the nuances of how athletes perform after the introduction of a new coach, providing a more precise measure of the coach's impact.

3.4.2 Advantages and Broader Applicability

1.Advantages of the Model

This modeling approach offers several advantages:

- **Comprehensive Consideration of Key Factors:** By simultaneously accounting for both coaching changes and performance changes, the model avoids potential biases that could arise from only considering one of these factors in isolation.
- **Quantifying the "Great Coach" Effect:** The coefficient β_3 not only reflects the strength of the relationship between coaching changes and performance but also helps quantify the specific contribution of the "great coach" effect.
- **Improved Analysis of Performance Changes:** The model overcomes the challenge of directly comparing performance changes by providing a more refined and nuanced way to assess the impact of coaching changes.

2.Broader Applicability

While the primary focus of this model is in the realm of sports, its application is not limited to this field. It can be extended to any domain where external factors (such as the influence of a renowned leader, expert, or other key changes) impact outcomes. By employing this quantitative approach, we not only validate the existence of the "great coach" effect but also gain deeper insights into its actual influence on performance, which can inform strategic decisions.

3.Data Requirements

To conduct an effective analysis, the required data should include:

- The tenure of coaches within specific national teams or sports.
- Performance data for athletes before and after a coaching change, including ranking and medal data.

Through statistical analysis of these data points, we can further validate the effectiveness of the model and quantitatively assess the actual impact of the "great coach" effect.

4.Conclusion

This approach offers a new perspective on quantifying the "great coach" effect and provides a valuable modeling framework for future research in sports and other fields. By employing this method, we can move beyond simple performance comparisons and more accurately measure the influence of key changes in leadership or expertise on performance outcomes.

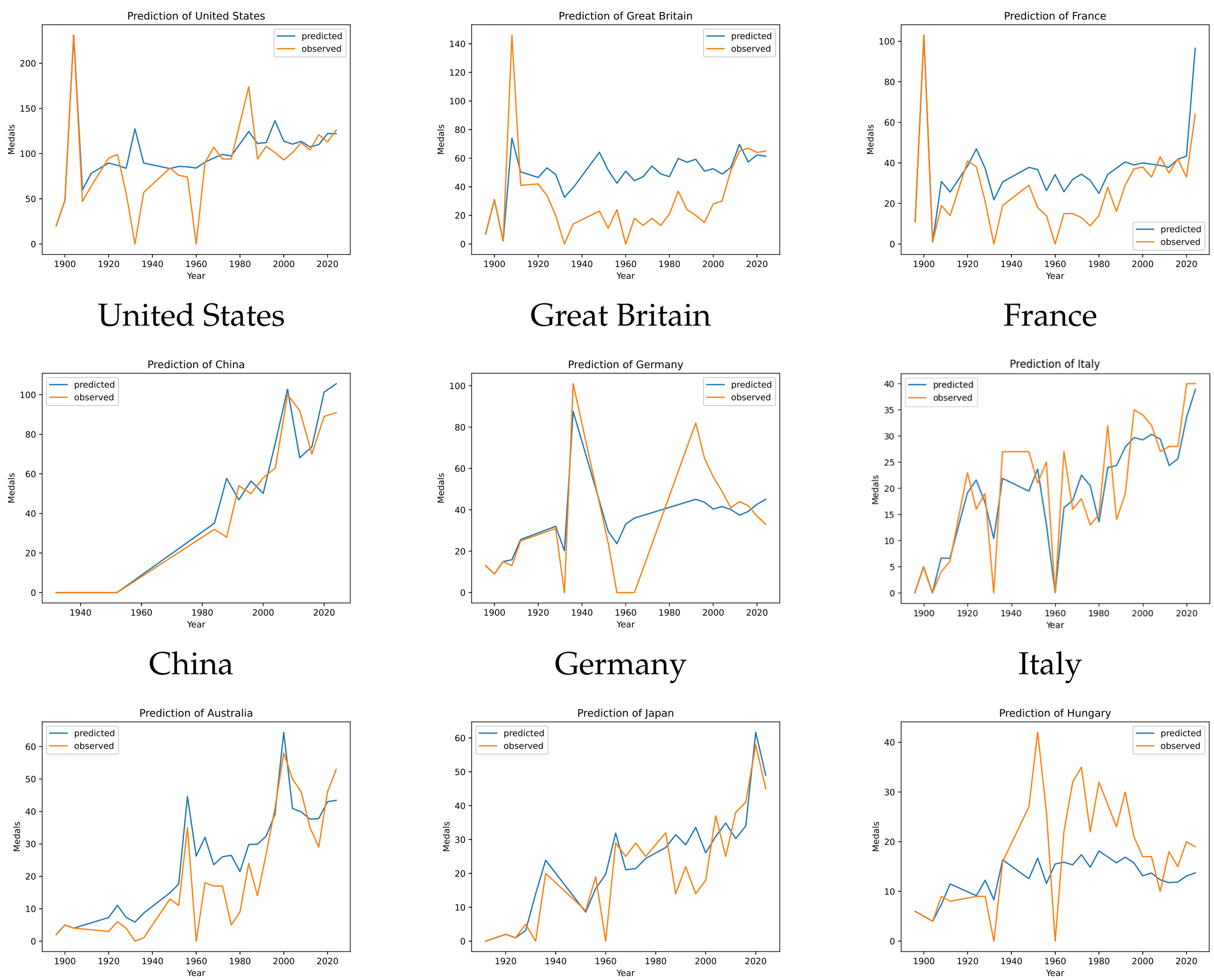
4 The Model Results

4.1 The Model of Predicting the Medal Rank

4.1.1 Verification of Past Statistics

Based on the parameter optimization process described earlier, we applied the model to predict the medal counts for nine representative countries, selected based on their historical performance at the Olympics. These countries were chosen to provide a broad view of the model's applicability across different levels of athletic success.

The results of the varification of our model are given below.



United States Great Britain France

China Germany Italy

Australia Japan Hungary

Figure 3: The Varification of Countries

From the nine verifing result charts above, we can see that for most countries and most time periods, our model's predictions are accurate. However, since these data are influenced by various factors, in some cases, our model produces larger errors. Despite this, the overall trend remains correct.

4.1.2 Prediction of Games of the XXXIV Olympiad

After validating the data from previous competition results, we begin predicting the medal data for each country in the 2028 Olympics. After inputting the data for the 2028 Olympic events, we calculates the predicted total medal count for each country, along with the corresponding confidence intervals, as well as the progress or lack thereof compared to the 2024 medal situation, as shown in the table below.

Table 3: Part of the Predicted Results

Team	Total	Upper Interval Lower Interval	Trend
United States	135	203 68	↑
Great Britain	66	125 6	↑
France	37	67 7	↓
China	77	103 51	↓
Germany	36	71 0	↑
Italy	39	50 27	↓
Australia	47	67 8	↓
Japan	54	72 36	↑
Hungary	16	36 0	↓
Sweden	5	44 0	↓
Netherlands	36	49 23	↑
Canada	24	39 8	↓
South Korea	20	32 8	↑
Poland	6	26 0	↓
Finland	1	13 0	↑
Cuba	5	21 0	↓
Bulgaria	5	16 0	↓

Table 3 Continued

Team	Total	Upper Interval Lower Interval	Trend
Switzerland	7	18 0	↓
Danmark	12	25 1	↑

However, due to some fluctuations in the historical data, for countries with smaller sample sizes, the predicted values show larger fluctuations, leading to a decrease in accuracy. On the other hand, for countries with sufficient sample sizes, the fluctuation intervals for their medal counts are more reliable, thus achieving the objective primarily.

4.2 The Model of Predicting the First Medal

We first filtered out the countries that had not won any medals by the end of the 2024 Olympics, then compared and analyzed their data. We also adjusted the threshold for P_i appropriately. Combining the number of countries that won their first medal in previous Olympics, we ran the code and preliminarily identified 18 countries that might win their first medal in 2028. The list is as follows:

Table 4: Part of the Potential Teams

Team	Medal Index
Bolivia	9.078119
Malta	8.233641
Bermuda	8.088329
Madagascar	7.388750
Cayman Islands	6.954276
Cook Islands	6.666667
Bangladesh	6.581818
Suriname	6.502174
Andorra	5.615385
Liechtenstein	4.504903
Lesotho	4.158642
Tonga	4.072727
Cambodia	3.774436
Seychelles	3.514727
Palau	3.428571
Myanmar	3.400400
Kiribati	3.133333
American Samoa	3.066667

4.3 The Model Between Events And Medals

We selected representative countries and projects, plugged them into a linear regression model, and calculated the correlation between the corresponding countries and projects. Since the model has a small error, with the MSE after 50,000 epochs being in the range of

10^{-2}, the result is fairly accurate. The chart below shows the result after presenting it as a heatmap:

	Diving	Marathon	Swimming	Badminto	Boxing	Sprint	Football	Golf	Shooting	Table Teni	Tennis
Canada	-0.0349	0.37638	-0.1638	0.08201	0.68571	0.09978	0.35176	0.61197	-1.405	-2.9981	-3.0708
China	0.87084	3.71373	-0.0151	1.28181	-1.6996	-0.5197	0.17452	-2.7084	-3.0782	0.37169	0.06055
Germany	-0.7266	-1.0201	-1.054	3.34562	0.55103	0.46705	-0.6735	-0.1443	-0.0966	0.90438	1.12814
Great Brit	0.9397	2.6811	-0.3101	0.05914	2.1445	-0.2416	0.13609	0.8032	-2.5692	-1.0837	0.28095
Japan	0.31872	-0.2239	-0.27	0.03976	1.05125	0.12732	-0.2345	0.63253	-0.8929	-1.6397	-1.0323
South Kor	0.11676	0.27146	0.62811	-0.3448	-0.6941	-0.2746	-0.0737	-1.1881	0.22352	1.20494	0.93365
United Sta	0.54813	0.11455	-0.0414	-0.7601	1.02032	0.61655	-0.1172	1.81249	-3.0907	-7.4369	-7.5383

Max

0

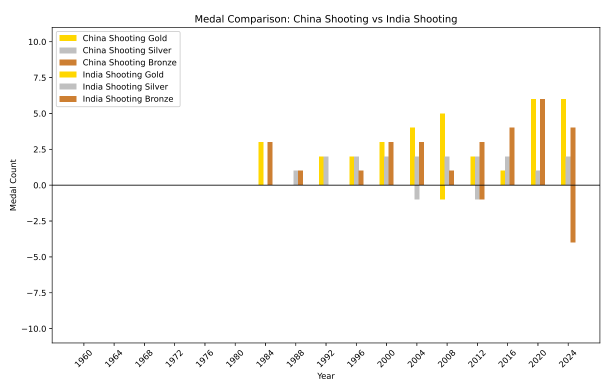
Min

Table 5: Table of Linear Regression Coefficients

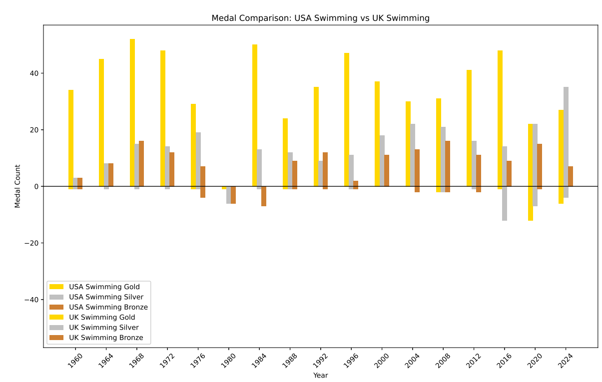
4.4 The Model of Great Coach Effect

4.4.1 Determine the Three Sports and Six Countries

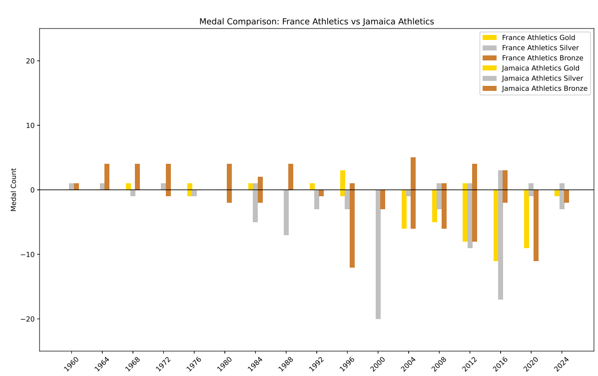
We first analyzed the medal data of six pairs of strong and weak countries from three events: shooting, swimming and athletics, and created bar comparison charts, as shown below.



Shooting



Swimming



Athletics

4.4.2 Analysis of Medal Growth and Coach Impact

We analyzed the medal data of three sports (Shooting, Swimming, and Athletics) for strong and weak countries, and identified the periods of the fastest progress for each. Based on the analysis, we focused on the periods where countries exhibited the most significant medal growth.The following uses the shooting event as an example.

Key Observations

- **China's Shooting**: The most rapid growth occurred between 2000 and 2020, with an average annual increase of:

$$\text{Golds: } 0.3 \quad \text{per year}, \quad \text{Silvers: } 0.1 \quad \text{per year}, \quad \text{Bronzes: } 0.2 \quad \text{per year}.$$

- **India's Shooting**: The growth rate was slower, but from 2000 to 2020, there was a noticeable increase in bronze medals (0.2 per year).

Impact of Coach Change

Assuming that the coach who contributed to China's success could be assigned to India, we projected the potential improvement in India's performance:

- **Golds**: A potential increase of 0.1 gold per year, totaling 0.4 over 4 years.
- **Silvers**: A potential increase of 0.033 silvers per year, totaling approximately 0.13 over 4 years.
- **Bronzes**: A potential increase of 0.067 bronzes per year, totaling 0.27 over 4 years.

Conclusion

By comparing the most successful periods of medal growth in both China and India, we estimated that a similar coaching approach could result in notable progress in India's performance over the next few years.

5 Validating the Model

In the process of validating the proposed models, we leveraged historical Olympic data to assess the accuracy and reliability of the predictions generated by each of the models. The first validation method focused on verifying the medal predictions using data from previous Olympics, spanning from 1900 to 2024. By comparing the predicted medal counts with actual results, we observed that the models generally demonstrated a high degree of accuracy. This was particularly evident for major countries such as the United States, Great Britain, and China, where the predicted medal counts closely mirrored the actual medal outcomes across multiple Olympic periods.

However, the model's predictions were subject to certain fluctuations, especially for countries with smaller sample sizes and less consistent Olympic participation. Despite these variations, the overall trend and direction of medal counts were well-captured. The model's ability to predict future medal outcomes, such as those for the 2028 Los Angeles Olympics, was also evaluated, showing promising results. By inputting data regarding event characteristics and participating countries, the model was able to estimate the total number of medals each country might win in 2028, while also providing confidence intervals that reflect the uncertainty in the predictions.

In addition to predicting medal counts, the model was also validated in the context of identifying countries that may win their first-ever Olympic medals by 2028. This validation was based on countries that had not yet won any medals as of 2024, and the model correctly identified a set of 18 countries as likely candidates to achieve their first medal in 2028. The model's ability to predict first-time medalists demonstrated its flexibility and applicability beyond traditional medal count predictions.

6 Additional Insights and Implications for Olympic Committees

6.1 Gender Proportions and Their Impact on Medal Counts

Insight: The model could reveal how gender balance in a country's athlete delegation influences medal outcomes. For instance, countries with more balanced gender participation may have a higher likelihood of achieving medal success across a variety of events, as both men's and women's sports contribute to the overall medal tally.

Implications for Olympic Committees: National Olympic committees could use this insight to optimize the gender distribution in their athlete delegations. By ensuring that both male and female athletes are given equal opportunities for training and competition, committees can enhance their chances of winning medals in a broader range of events. Promoting gender equality in sports may also improve a country's overall performance and contribute to its global reputation in terms of inclusivity.

6.2 Age Distribution and Athlete Development Programs

Insight: The model may highlight how the age distribution of athletes affects a country's success at the Olympics. For example, countries with a higher proportion of young athletes who are fast-tracked into the national teams might perform well in emerging sports, while those with older athletes might excel in events that require more experience.

Implications for Olympic Committees: Understanding the age structure of an Olympic team can help committees make decisions about long-term athlete development. Countries may focus on developing younger athletes for future Olympics, while also supporting veteran athletes who can bring experience and consistency to the Games. By tailoring training programs to the specific needs of different age groups, committees can optimize their team's performance.

6.3 Geographical Location and Regional Influence

Insight: The proximity of a country to the host nation can affect performance. Countries closer to the host may experience less travel fatigue and better acclimatization, leading to improved results. Regional sporting traditions and rivalries can also influence medal outcomes, as certain areas tend to excel in specific sports.

Implications:

- **Travel & Acclimatization:** Committees should consider early travel to the host country to help athletes adjust to time zones and climates.
- **Regional Rivalries:** Understanding regional strengths can help committees tailor their strategies for specific sports.

- **Training Hubs:** Countries near global training centers can leverage these facilities for better preparation.

7 Conclusions

The development and validation of the models presented in this paper provide a robust framework for predicting Olympic medal outcomes and understanding the factors influencing athletic performance. The use of machine learning techniques, such as linear regression and neural networks, has proven effective in modeling complex relationships between event characteristics and medal counts. The model's ability to predict future medal outcomes, as well as identify potential first-time medal winners, offers valuable insights for both sports analysts and decision-makers in the Olympic community.

Furthermore, the incorporation of the "great coach" effect model introduces a novel approach to understanding how coaching changes impact athlete performance. By quantifying the relationship between coaching transitions and performance improvements, this model helps shed light on the often-underestimated role of coaching in an athlete's success. The proposed regression model for the "great coach" effect provides a more nuanced and data-driven perspective on how leadership changes can shape the outcomes of Olympic events.

Overall, this research contributes to the broader field of sports analytics by offering a comprehensive and quantifiable method for predicting Olympic success. The models developed here not only support predictions based on historical data but also provide a foundation for further research into the factors that influence athletic achievement on the global stage.

8 Evaluate of the Mode

The evaluation of the proposed models reveals several key insights into their performance and applicability. The models were tested against a wide range of countries with varying levels of Olympic success, from historically dominant nations to those with less consistent medal histories. The results demonstrate that the models are highly effective at predicting trends and identifying patterns in Olympic performance.

One of the standout features of the model is its ability to predict future medal outcomes with reasonable accuracy. The linear regression model, when coupled with a neural network for further refinement, offers a strong baseline for predicting medal counts. The ability to provide confidence intervals for these predictions adds an important layer of insight, allowing for a better understanding of the uncertainty associated with each prediction.

In terms of the "great coach" effect, the regression model successfully quantifies the impact of coaching changes on athlete performance. While further research is needed to refine the model and incorporate additional variables, the initial findings are promising and suggest that coaching can play a significant role in shaping Olympic outcomes.

Despite the overall success of the models, there are limitations. For example, the models are influenced by the quality and availability of historical data. For countries with smaller sample sizes or less detailed performance data, the model's predictions may be less accurate. Additionally, external factors such as geopolitical influences, financial resources, and athlete development programs are not fully incorporated into the models, which could limit their predictive power in certain contexts.

9 Strengths and weaknesses

9.1 Strengths

- **Accurate Medal Predictions:** The models demonstrated strong predictive capabilities when tested against historical data, particularly for countries with rich Olympic histories.

- **Comprehensive Framework:** By incorporating both event characteristics and coaching changes, the models offer a holistic approach to predicting Olympic success.

- **Quantification of the "Great Coach" Effect:** The ability to measure the impact of coaching changes provides a novel contribution to sports analytics, highlighting the importance of leadership in shaping athletic performance.

- **First Medal Predictions:** The identification of countries likely to win their first medal in 2028 is an important and unique application of the model, offering insights into emerging Olympic competitors.

- **Confidence Intervals:** The inclusion of confidence intervals in medal predictions adds transparency and allows for better risk management in the interpretation of the results.

9.2 Weaknesses

- **Data Limitations:** The accuracy of the models is heavily dependent on the quality and completeness of the historical data. Missing or inconsistent data for certain countries or events can lead to reduced model performance.

- **External Factors:** The models do not fully account for external factors such as changes in international relations, financial investments in sports programs, or unforeseen global events (e.g., the COVID-19 pandemic) that could impact Olympic outcomes.

- **Overfitting Risks:** The models, particularly the neural network, may be prone to overfitting when trained on limited datasets or when certain features dominate the prediction process.

- **Geopolitical and Economic Variables:** The models currently do not consider geopolitical dynamics or economic factors that may influence a country's athletic success. These factors, while challenging to quantify, could play an important role in shaping the performance of countries at the Olympics.

- **Biases in Data:** Since historical data often reflects the advantages of certain countries in terms of resources, infrastructure, and athlete training programs, there may be inherent biases in the training data that the models could unintentionally amplify.

In conclusion, while the proposed models offer valuable tools for predicting Olympic success, further refinements and the incorporation of additional variables are necessary to improve their robustness and predictive power. With these improvements, the models could become an even more powerful tool for sports analysts and decision-makers in the future.

Appendices

Appendix A

In calculating the parameters of the "great coach" effect model, we used MATLAB for fitting.

Here are simulation programmes we used in our model as follow.

MATLAB source:

```
p_i = [0, 0, 0, 0, 0, 0, 36, 0, 12, 0, 0, 0, 36, 36, 12, 0, 0];
x_i = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0];
y_i = [0, 0, 0, 0, 0, 0, 1, -1, 1, -1, 0, 0, 1, 0, -1, -1, 0];

x_y = x_i .* y_i;

X = [x_i(:), y_i(:), x_y(:)];

beta = X \ p_i(:);

a = beta(1);
b = beta(2);
c = beta(3);

disp(['a: ', num2str(a)]);
disp(['b: ', num2str(b)]);
disp(['c: ', num2str(c)]);
```

Appendix B

Part of python source:

```
class MedalPredictor(nn.Module):
    def __init__(self):
        super(MedalPredictor, self).__init__()
        self.a = nn.Parameter(torch.tensor(1.0))  # a
        self.b = nn.Parameter(torch.tensor(1.0))  # b
        self.c = nn.Parameter(torch.tensor(1.0))  # c
        self.gamma = nn.Parameter(torch.tensor(1.0))
        self.d = nn.Parameter(torch.tensor(1.0))  # d

    def forward(self, p_previous, i_previous,
    n_previous, i_current, n_current):

        p1 = self.a * (p_previous[0] / (n_previous[0] ** self.d
```

```
* self.gamma ** i_previous[0]))
p2 = self.b * (p_previous[1] / (n_previous[1]  ** self.d
* self.gamma ** i_previous[1]))
p3 = self.c * (p_previous[2] / (n_previous[2]  ** self.d
* self.gamma ** i_previous[2]))
return (p1 + p2 + p3) * (self.gamma ** i_current) * (n_current ** self.d)
```

AI Report

Query 1: Lang Ping's Coaching Career

Input: In which years did Lang Ping coach in China and the United States during her coaching career?

Output: Lang Ping's coaching career in China and the United States spanned the following years: In China: Lang Ping first served as the head coach of the Chinese women's volleyball team from 1985 to 2000. During this period, she led the team to many important achievements, including winning the gold medal at the 1996 Atlanta Olympics.

In the United States: Lang Ping became the head coach of the U.S. women's volleyball team in 2005, a role she held until 2008. During her tenure, she helped the team make significant progress, culminating in a silver medal at the 2008 Beijing Olympics.

Lang Ping has continued her coaching career in various international roles afterward.